

The Trip Economy: Which Foul-Drawing Skills Travel?

Introduction

In 2025-26, a two-shot trip at league conversion yielded 1.57 expected points against 1.09 for the average field attempt. Yet foul drawing is measured coarsely. Free-throw rate treats all trips alike, and possession arithmetic approximates trip counts with the 0.44 coefficient. Recent work folded the free throws earned on fouled shots into shot value, and existing tools can count the rest of the trip economy. We ask how players actually get to the line and which channels are repeatable player skills rather than circumstance.

Methods

We reconstruct every regular-season NBA free-throw trip, meaning the free throws awarded from a single non-technical foul, from official play-by-play across the three most recent seasons (2023-24 through 2025-26, 3,690 games). We link each trip to its recorded causing foul and classify trips by foul type under documented rules into eight classes in two tiers. Attempt-equivalent trips replace a field-goal attempt as the possession's scoring attempt (two- and three-shot shooting fouls, bonus). Add-on trips stack points on a possession that already stands (and-ones and rarer classes). Trip counts are exact rather than estimated: reconstructed free-throw makes and attempts reconcile with the official box score for every player-game and with an independent season-totals source for every one of 1,723 league player-seasons, yielding 88,347 trips with zero discrepancies. The dataset and pipeline are public.

Results

The trip economy varies structurally across the 284 players with 300 or more field-goal attempts in 2025-26 (Table 1); James Harden leads in three-shot fouls drawn (3.8 per 100). Channels persist differently year over year (422 player-season transitions; the pattern holds in both season pairs): two-shot shooting fouls $r = .87$, three-shot fouls $.76$, and-ones $.75$, bonus $.58$, residual classes $.31$. Two-shot fouls persist near their within-season (split-half) reliability of $.89$ to $.92$; bonus trips fall short of a lower $.65$ to $.78$, noisier to measure and less stable (Figure 1). Across 54 mover and 284 stayer transitions (midseason changes excluded), movers showed lower persistence in every main channel: all trips $.74$ against $.87$ (player-cluster bootstrap 95% interval for the gap $+.02$ to $+.26$). The bonus gap was largest ($.62$ against $.40$) but its interval spans zero ($-.05$ to $+.58$), so channel-specific differences remain uncertain. The conventional 0.44 coefficient tends to mismeasure the same players every year: its true value spans $.277$ to $.481$, persists at $r = .53$, and moves True Shooting by a median 0.15 and up to 1.5 percentage points (Table 1).

Conclusion

Two-shot shooting-foul generation is the most persistent of the four main channels, for stayers and movers alike; non-shooting bonus generation is the least. The split motivates channel-specific projections, with portability differences across team changes still uncertain, and exact trip accounting corrects a persistent, player-shaped error in efficiency metrics.

Channels persist differentially; non-shooting bonus trips persist least

Each dot: the correlation between a player's rate in one season and the next, over player-season pairs where he stayed with one team (blue, $n = 284$ transitions) or changed teams (orange, $n = 54$; 84 pairs with a midseason move excluded). Whiskers: player-cluster bootstrap 95% intervals; the text under each row is the interval for the blue-orange gap. Gray tick: within-season (split-half) reliability, 2025-26.

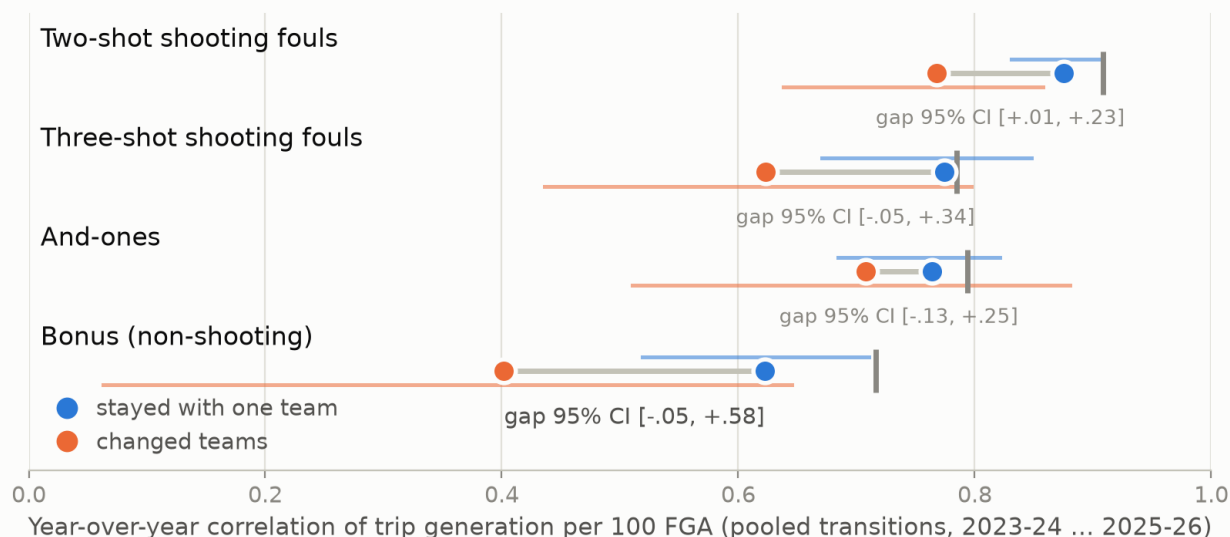


Figure 1. Year-over-year persistence by channel for stayer vs mover transitions, with player-cluster bootstrap intervals, against each channel's within-season reliability; pooled over 2023-24 ... 2025-26.

Table 1. The trip taxonomy, measured. League trips count every 2025-26 trip; the rate distribution and split-half reliability are over the 284 players with ≥ 300 field-goal attempts in 2025-26; year-over-year r is pooled over 2023-24 ... 2025-26.

Channel	Tier	League trips (2025-26)	Median /100 FGA (p10–p90)	Year-over-year r	Split-half reliability
Two-shot shooting fouls	attempt-equivalent	18,846	8.1 (3.1–14.0)	0.87	0.91
Three-shot shooting fouls	attempt-equivalent	933	0.2 (0.0–1.1)	0.76	0.78
And-ones	add-on	5,860	2.6 (0.9–4.5)	0.75	0.79
Bonus (non-shooting)	attempt-equivalent	4,760	1.8 (0.8–3.7)	0.58	0.72
Other add-on (flagrant, away-from-play, transition, clear path)	add-on	549	—	0.31	—
All trips	—	30,948	—	0.86	0.95

Note. The true attempt-equivalent coefficient — the exact replacement for the conventional 0.44 — has median 0.427 across qualified players (p10–p90 0.385–0.455, full span 0.277–0.481) and persists year over year at $r = 0.53$.